Chapter 4

Network Layer

- We are studying about the network layer of the internet.
- Provides best effort service only.

Network layer protocol is

i) connectionless

ii) stateless

iii) unreliable.

No guarantee is provided about

i) congestion

ii) Bandwidth

iii) Timing

iv) In order delivery.

Three protocols run at - Network Layer

1) Routing Protocol

2) ICMP (error checking).

3) Internet Protocol

Packet handling diagram format Addressing conventions.

IP V4 Header

32 bits

Version 4 bits	Header len 4 bits	Type of service 8 bits	Total length 16 bits.
identification 16 bits.		flags. 3 bits	offset of fragmentation 13 bits
Time to live 8 bits	Protocol at upper layer. 8 bits		Header checksum 16 bits.
Source IP			
Destination IP.			
options. 0 to 40 bytes.			
Data. without IP header. (UDP/TCP segment			

Max IP header = 60 bytes.

Min IP header = 20 bytes. (first 5 fields are mandatory) each field is of 4 bytes (32 bits).

- whatever is written in total length and header length is the length in $\boxed{\text{bytes}}$
- Header length is scaled by a factor of 4.
- Total length is $\boxed{\text{length of data + length of header.}}$

- minimum value in total length field is 20 bytes and max is 2^{16} = 65536 bytes.

FRAGMENTATION

- MTU is the maximum transmissible unit at the data link layer.
- value of MTU is the maximum length of data (network layer datagram) that can be kept in data / payload field of data link layer frame.
- Datagram is divided into fragments on the basis of MTU
- Each fragment is treated as independent datagram by the routers.
- A fragment can be further subdivided into fragments as well.
- All fragments are reassembled at destination system as there can be multiple / different routes taken by each fragment.

Example

4000 byte Datagram

Header = 20 bytes.

Actual Data = 3980 bytes.

MTU = 1500 bytes.

Header = 20 bytes. Data = 1480 bytes. → is a multiple of 8 here

↓

each ~~frame~~ fragment will have to carry header since each fragment is a datagram.

Now, lets divide into the fragments

length	ID	fragflag	offset
1500	x	0	$\frac{1480}{8} = 185$
1500	x	1	

Maintained in accordance with actual data length given

	length	ID	fragflag	offset
1	1500	x	1	0 (0 – 1479 has been sent)
2	1500	x	1	$\frac{1480}{8} = 185$ (1480 – 2959 has been sent)
3	1040	x	0	$\frac{1480 \times 2}{8} = 370$. (2960 – 3980 has been sent).

Skip headers, look in terms of data only.

In ① we have sent ~~3980~~ 1480 bytes of actual data.

Data sent by ① = 1480 bytes.

Data remaining = 3980 − 1480 = 2500

Total data sent = 1480 bytes to destination.

offset (starting byte) = 0.

In ②

Data sent by ② = 1480 bytes.

Total Data sent to dest = 1480 + 1480 = 2960

Data remaining = 3980 − 2960 = 1020

offset = $\frac{1480}{8}$ = 185.

In ③

Data sent by ③ = 1020 bytes.

Total Data sent to dest = 1480 + 1480 + 1020 = 3980.

Data remaining = 3980 − 3980 = 0.

offset = $\frac{2960}{8}$ = 370

Note

- Each fragment inherits its parent ID.
- frag flag 0 → no fragments would be followed

  frag flag 1 → more fragments to be followed.

→ Example 2

MTU = 1520

Data = 4000.

Header = 20

Actual Data = 3980.

Actual MTU Data = 1500 (not a multiple of 8).

Now,

$\frac{1500}{8} = 187.5$

find number closest to 1500 which must be multiple of 8.

ignore after decimal part in 187.5 = 187

$187 \times 8 = \boxed{1496}$ closest multiple of 8.

Actual MTU Data = 1496 } only this changed.

Actual Data = 3980 } Same

Header = 20

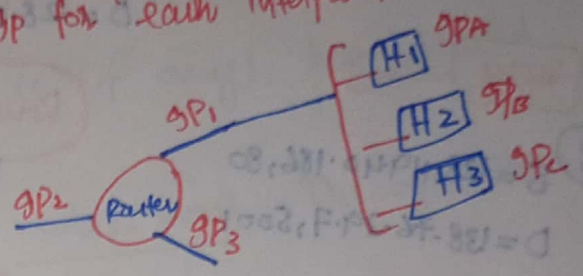
length	ID	flag	offset
1510	x	1	0
1510	x	1	187
1008	x	0	374

→ Data sent = 1496
Rem = 3980 − 1496 = 2484

→ Data sent = 2992
Rem = 3980 − 2992 = 988

→ IP ADDRESSES

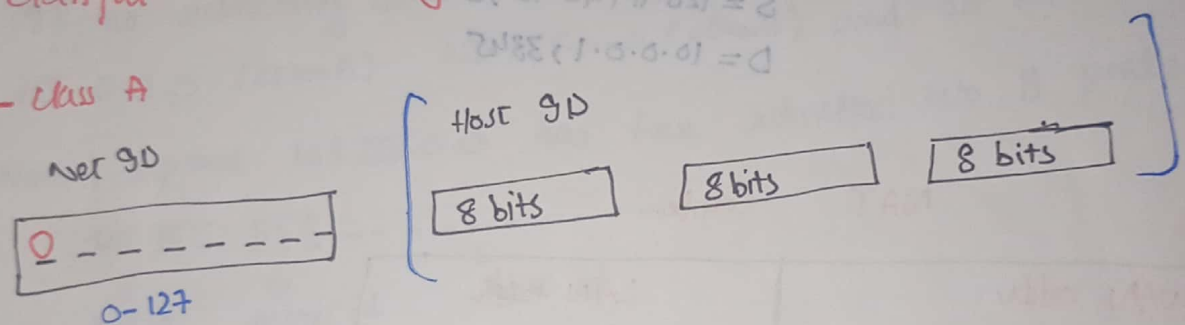
- Always associated with the interface not with device / router.
- Interface is the link of connection between device / router and physical link.
- In terms of internet IP address is given to NIC (network interface card) of host. Routers always have multiple interfaces. (unique IP for each interface).



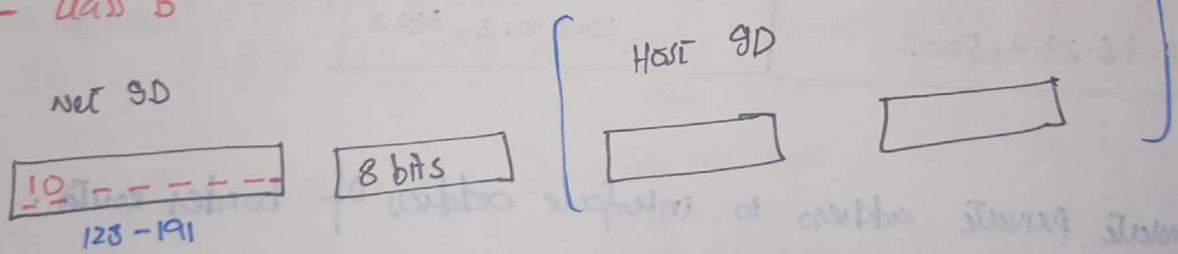
- Two types of IP addressing
  - → classful
  - → classless

- Classful IP Addressing

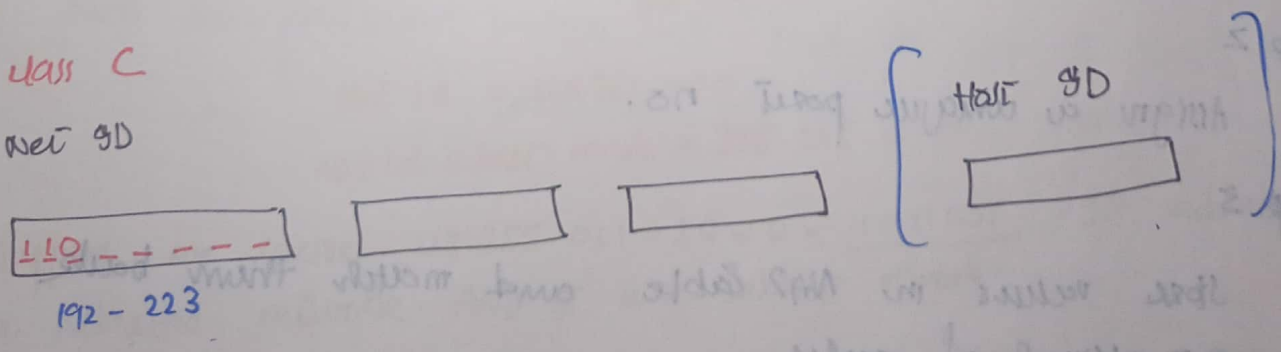
— class A



— class B

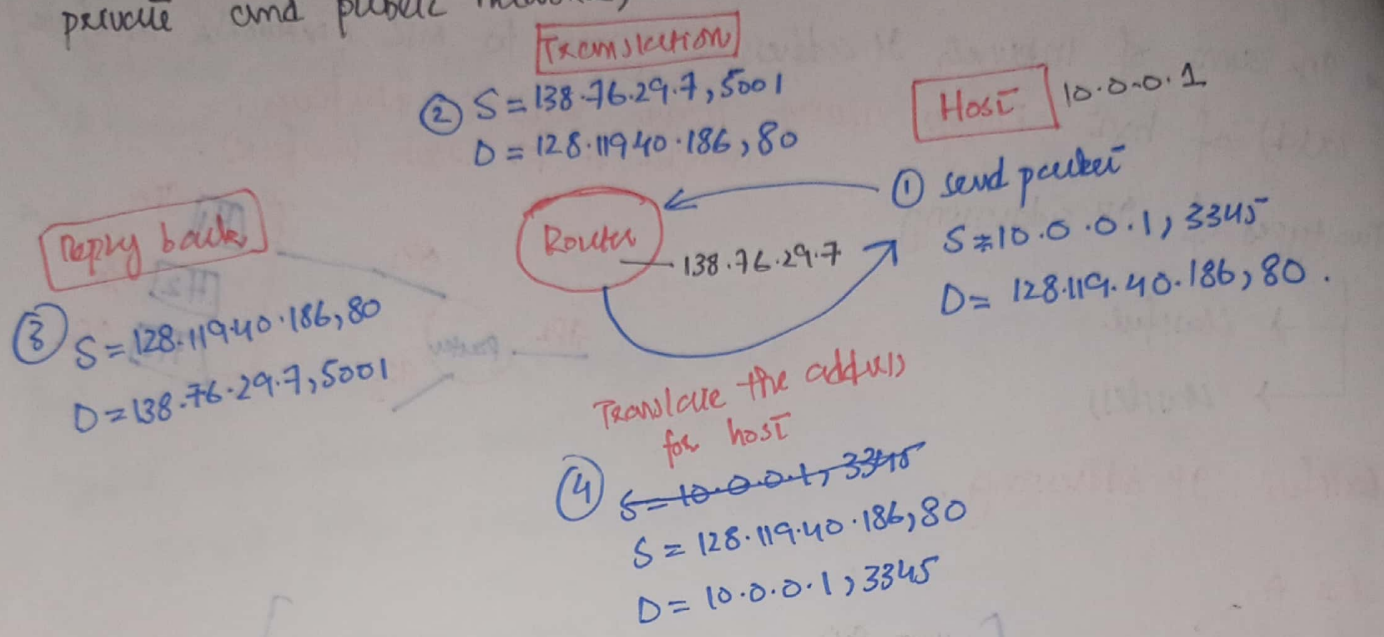


- class C



→ NAT (Network Address Translation).

- Private IP addresses allocated to hosts within a network.
- IP address is assigned to the local network only.
- Implemented through boarder access router (router separating private and public network).



NAT Table

WAN addr	LAN addr
138.76.29.7, 5001	10.0.0.1, 3345
138.76.29.7, 5002	10.0.0.2, 8423

- Step 1
  Translate private address to interface address of border router.
- Step 2
  Assign a unique port no.
- Step 3
  Store values in NAT Table and match them back at arrival of reply.

→ subnet

- host ID portion is further subdivided into networks.
- ~~sub~~ subnets always formed from host id portions, netid is always fixed.

→ Subnet Mask.

- Process of extracting netid or subnet id from an IP address.
- set net id all 1s, set subnetid all 1s (if any), keep hostid bits all 0s.
- Do ANDING between IP address and subnet mask. Through this process, we get net ID or subnet ID.
- If no subnetting done then it is easy to form subnet mask. 255.0.0.0 (class A) 255.255.0.0 (class B) and so on.
- Now, suppose 128.255.0.0 has been subnetted into 8 parts. Then

128.255. SSS - - - - - - - - - - - - -

↓ netid ↓ subnet id ↓ host id.

Subnet mask = 255.255.11100000.00000000

= 255.255.224.0 same subnet mask for 8 subnets in this case.

- But different subnet ids id1 = 128.255.000 . . . . .

128.255.001 . . . .

128.225.010 . . . .

Net id = 128.255.000 . . . .

netid subnet mask = 255.255.0.0.

Suppose, we receive 128.255.00111000.11011101 IP address on a subnetted network. AND it with subnet mask.

(Mask) 11111111.11111111.11100000.00000000

(IP) 10000000.11111111.00110000.11011101

10000000.11111111.001 00000.00000000

Net ID subnet2 host 0 (all 0s not assigned)

FLSM (fixed length subnet mask)

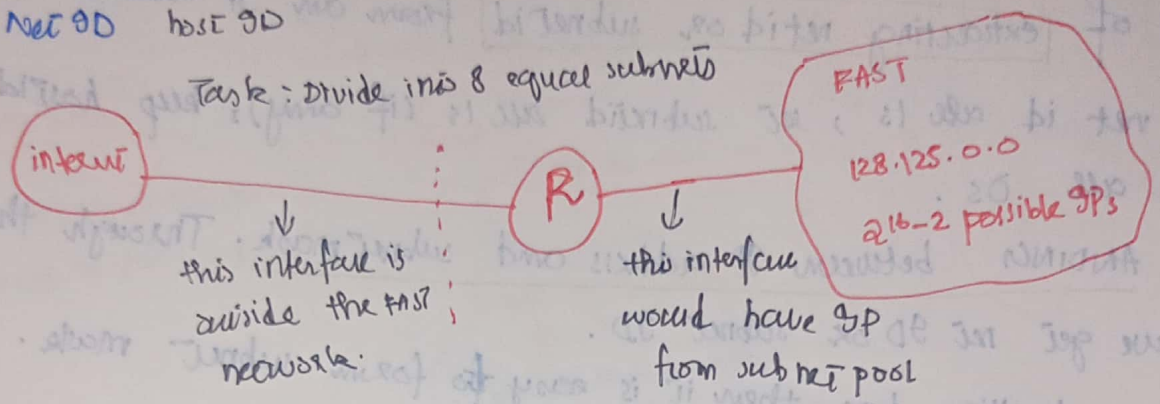
can either be classful or classless.

- classfull first

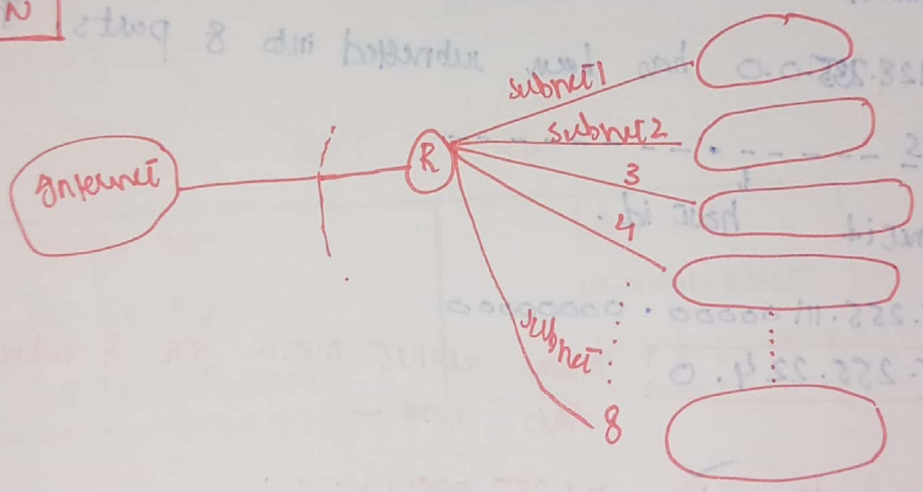
128.125.0.0 → class B.

Net ID host ID

Task: Divide into 8 equal subnets



SOLUTION



- Net ID would be fixed, steal from host ID.

We have 16 bits of host ID. without subnetting, we have $2^{16}-2$ ~~subnets~~ hosts in the network.

subnets $= 8 = 2^3$

Each subnet will have $= \frac{2^{16}}{2^3} = \underline{2^{13} \text{ hosts}}$.

subnet id bits to be used $= 16 - 13 = \underline{3 \text{ bits}}$.

128 . 125 . S S S - - - - - - . - - - - - - - - - -

Net id of 16 bits.

subnet ids of 3 bits.

host id of 13 bits.

128.125.SSS . - - - - - . - - - - - - - - -

000
001
010
011
100
101
110
111

8 subnets formed now.

Subnet address 1 = 128.125.0.0

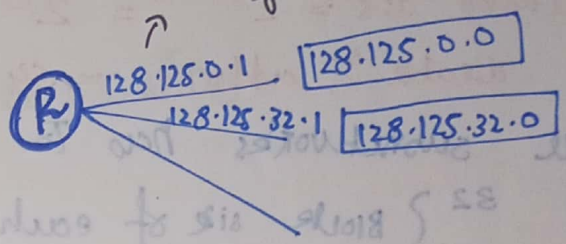
" 2 = 128.125.32.0

" 3 = 128.125.64.0

⋮

8 = 128.125.224.0

Router interface usually gets first subnet address.



- All 0s and 1s host id bits are unuseable in subnets as well.

Here Net ID 128.125.0.0 and first subnet id 128.125.0.0 are same but their corresponding subnet masks would be different.

255.255.0.0 for net id

255.255.11100000.00000000

255.255.244.0 for subnet id 1

subnet masks.

→ Variable length subnetting mask (VLSM).

a.b.c.d / n → no. of network or subnet id bits.

→ / indicates classless.

Block size (available IP addresses) = (2^{32-n}) → all 0s and 1s would not be assigned (of host ID).

VLSM is only for classless addressing.

Net ID → 17.12.40.0/26 (organization has been allocated this IP address)

Available IPs = $2^{32-26} = 2^6 = 64$ (6 bits of host)

Useable IP addresses = 64-2 = 62

Three subnetworks has to be made.

A 32
B 16
C 16

} Block size of each subnet.

17.12.40.00000000/26
net id | host id

to

17.12.40.00 111111/26
net | host

→ Block condition before subnet
All IP addresses assignable except first and last.

- we fix bit in host id part one by one, so

17.12.40.00 [0] 00000
↓
fixed

17.12.40.00000000/27
net | sub | host

Block 1 :

17.12.40.00 0 11111/27

$2^5 = 32$ addresses first and last is not assignable.

A needed block of 32 so assign this block to A.

17.12.40.000 - - - - -

Now make the fixed bit 1 (I have missed 17.12 everywhere).

17.42.001 0 0 0 0 0/27

⋮

Block 2.2

17.42.001 1 1 1 1 1/27

$2^5 = 32$ addresses but requirement is lower than that.

further subnet the block B now.

17.42.001 (0) 0 0 0 0/28

Block 2.1

Dept B

17.42.001 (0) 1 1 1 1/28

$2^4 = 16$.

Assign the block to B.

Now make the fixed bit 1

17.42.001 (1) 0 0 0 0/28

Block 2.2

Dept C

17.42.001 (1) 1 1 1 1/28

→ Assign this to C.

$2^4 = 16$

1) What are the subnet masks?

2) Make the bits written after [1] one.

3) 18.17.42.00111101/28 (make the first 28 bits 1 and rest would be 0).